

Critical Care Case 2: Myasthenic Crisis

Section 1: Case Summary

Scenario Title:	Myasthenic Crisis
Keywords:	Respiratory failure, difficult airway
Brief Description of Case:	57-year-old female with known history of myasthenia gravis and thymoma is admitted to hospital 48 hrs prior for CAP and started on Ceftriaxone and Azithromycin. Over the last 24 hrs, she has developed progressive weakness and bulbar symptoms. RT was paged to assess vital capacity, and finds her in respiratory distress, at which point the Rapid Response Team (sim team) is paged. The resuscitation team is expected to recognize and manage myasthenic crisis with respiratory failure, with consideration for airway management with an anterior mediastinal mass. The patient will require intubation and pharmacotherapy for myasthenic crisis. Additionally, the team is expected to update the family member, and inform them of medical error.

Goals and Objectives	
Educational Goal:	Understand the management of myasthenic crisis with respiratory failure.
Objectives: (Medical and CRM)	1. Assess the patient with respiratory failure due to neuromuscular weakness. 2. Identify and manage a difficult airway due to anterior mediastinal mass. 3. Utilize appropriate medical therapy for myasthenic crisis. 4. Identify the risks and benefits of transporting a patient to another location vs managing immediately. 5. Identify the need for additional support from consulting specialists. 6. Disclose medical error.
EPAs Assessed:	C1 Resuscitating and coordinating care for critically ill patients. TP3 Managing complex interpersonal interactions that arise during the course of a patient's care.

Learners, Setting and Personnel	
Target Learners:	☐ Junior Learners ☒ Senior Learners ☐ Staff
	☒ Physicians ☒ Nurses ☒ RTs ☒ Inter-professional
	☐ Other Learners:
Location:	☒ Sim Lab ☐ In Situ ☐ Other:
Recommended Number of Facilitators:	Instructors: 1
	Sim Actors: 3-4
	Sim Techs: 1

Scenario Development	
Date of Development:	June 6, 2019
Scenario Developer(s):	D. Piquette
Affiliations/Institutions(s):	University of Toronto
Contact E-mail of Editors:	Sophie.ramsden@medportal.ca; Sameer.sharif@medportal.ca
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Revised By:	S. Ramsden, S. Sharif
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Miranda Ford			Age: 57	Gender: Female	Weight: 75 kg
Reason for Referral: Respiratory Failure					
Temp: 37.0	HR: 105	BP: 130/65	RR: 25	O ₂ Sat: 93%	FiO ₂ : 4L NP
Cap glucose: 6.7			GCS:15 (E4 V5 M6)		
Referral Information: 57-year-old female with history of myasthenia gravis currently admitted for pneumonia presents in respiratory distress. RT was paged to assess the patient and paged you urgently because the patient appears unwell.					
Allergies: none					
Past Medical History: Myasthenia Gravis (diagnosed recently) Anterior Mediastinal Mass (likely thymoma), to be resected in a few weeks			Current Medications: Mestinon 90 mg QID Ceftriaxone 2g IV Q24H Azithromycin 500 mg IV Q24H		

Section 2B: Extra Patient Information

A. Further History	
<i>History provided by RT:</i> Mrs Ford is a 57-year-old female with a history of myasthenia gravis that RT was paged for to assess vital capacity. She is admitted under the internal medicine service for pneumonia, and they were concerned that she has had rapid progression of her myasthenic symptoms in the last 24 hours. RT assessed her vital capacity at 1.2L.	
<i>History obtained from chart:</i> Mrs Ford was admitted to hospital 48 hours ago for community-acquired pneumonia, and treated with ceftriaxone and azithromycin. Her home dose of mestinon was continued. At the time of her admission, her exam was remarkable for O ₂ sat of 93% on 4L, right lower lobe crackles, and a fever. However, her neurologic exam at the time was normal. Chart review should also reveal her anterior mediastinal mass, awaiting surgery.	
<i>History obtained from patient:</i> Over the last 24 hours, her myasthenic symptoms have rapidly progressed with general weakness in 4 limbs and bulbar symptoms, including poor swallowing and intermittent diplopia.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: normal	Neuro: Power 3+/5 x 4, weak cough, difficulty lifting head from pillow.
Resp: shallow breathing, decreased AE to bases, b/l crackles / upper airway secretions.	Head & Neck: Normal voice
Abdo: soft, nontender nondistended	MSK/skin: normal
Other: Appears anxious. Says she feels "like suffocating". Speaks in short sentences.	



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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
ECG leads NIBP Cuff Pulse oximeter Peripheral IV Airway equipment (nasal prongs, non-rebreather, BVM, suction) BiPAP Intubation supplies (bronchoscope, VL, DL, bougie, LMA) Defibrillator and pads
C. Required Medications
Fluids Induction agents Paralytics Vasopressors ACLS medications (available)
D. Moulage
N/A
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed ☐ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> A – Speaking in short sentences. B – 4L NP. Shallow breathing, bilateral crackles / upper airway secretions, decreased air entry to bases. C – Warm to touch, pulses x4. Abdo soft. D – GCS 15. PERRLA. Power 3+/5 x4. Weak cough, difficulty lifting head off pillow. E – Temperature 37. If deep suctioning or awake intubation is attempted, patient becomes agitated, making it difficult to complete these tasks.

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Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
RT	Gives additional history (see information above). Provides ABG at 8 min. May assist with airway and breathing as directed by team.
Patient's husband	Arrives at the end of the scenario, wanting to know what happened and why his wife deteriorated so quickly. Will become upset, especially if anterior mediastinal mass was not taken into account in her airway management.
RN	Assists with starting IVs and administering medications.

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: sinus HR: 105 BP: 130/65 RR: 25 O ₂ SAT: 93% on 4L NP T: 37.3°C GCS: 15	<i>Patient is alert and anxious, taking shallow breaths. She speaks in short sentences.</i>	<u>Expected Learner Actions</u> ☐ Take a history from the patient, RT, and chart ☐ Examine the patient ☐ Recognize that this is myasthenic crisis ☐ Call for help – anesthesia, ICU staff	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - elevating head of bed => patient feels a little better - increased FiO ₂ => Sat up to 95% <u>Triggers</u> <i>For progression to next state</i> - 5 minutes elapsed -> 2. Desaturation	- prompt team with patient's chart, if not reviewed - anesthesia and ICU take team's call, but are too busy to come assist
2. Desaturation Rhythm: sinus HR: 105 BP: 130/65 RR: 25 O ₂ SAT: 87% on 4L NP T: 37.3°C GCS: 15	<i>O₂ sats drift to 87%</i>	<u>Expected Learner Actions</u> ☐ Discuss an airway plan ☐ Discuss the benefits of NIV vs PPV ☐ Discuss risks & benefits of transporting to another location vs immediate management ☐ Attempt suctioning ☐ Page anesthesia for help ☐ Discuss deferring intubation to OR setting for rigid bronchoscopy	<u>Modifiers</u> - suction => improved sats, but patient becomes agitated, uncooperative - BiPAP / Optiflow => initial improvement in O ₂ sat to 92%, followed by recurrent desat - if team decides to transport patient => terminal desaturation <u>Triggers</u> - Decision to intubate -> 3. Definitive Airway	- inform team patient becomes too agitated to continue with suction if attempted - At 8 minutes, if not moved onto the next state, provide ABG performed by RT before team's arrival - if team does not proceed to intubation at 10 minutes, continue to decrease O ₂ saturations progressively more rapidly - anesthesia is busy/unavailable if paged



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3. Definitive Airway Rhythm: sinus HR: 120 BP: 130/65 RR: 32 O ₂ SAT: 85% on NP/BiPAP/Optiflow T: 37.3°C GCS: 15	<i>Progressive desaturation and increased agitation</i>	<u>Expected Learner Actions</u> ☐ Recognize the possibility of a difficult airway, given the anterior mediastinal mass ☐ Communicate airway plan to the team, including a backup plan ☐ Trial awake intubation ☐ Use induction agents ☐ Position HOB up for intubation ☐ Consider medications for myasthenic crisis (IVIG/plasmapheresis, high-dose steroids, stop pyridostigmine) ☐ Call neurology for advice	<u>Modifiers</u> - HOB down to position patient for ETI => Sat and BP drops by 20% - Trial of awake intubation => patient is combative, shaking head - full dose sedation => post-intubation hypotension requiring pressors - light sedation in semi-seated position => intubation successful <u>Triggers</u> - intubation completed and next steps discussed -> 4. Conversation with Spouse	- inform team manikin becomes too agitated to continue with awake intubation if attempted
4. Conversation with Spouse	<i>Spouse walks in when intubation is completed, demanding to know what happened to his wife.</i>	<u>Expected Learner Actions</u> ☐ Explain to spouse what has happened ☐ Disclose that an error occurred in administering azithromycin	<u>Modifiers</u> - team informs spouse what happened honestly and compassionately -> he is upset but reasonable - team omits medical error or has not accounted for mediastinal mass in intubation -> he becomes more upset <u>Triggers</u> - End case here	If the team has not accounted for the anterior mediastinal mass, the spouse should specifically bring it up.

Critical Care Case 2: Extra Documents

Appendix A: Laboratory Results

<u>ABG</u> pH 7.34 pCO ₂ 47 pO ₂ 85 HCO ₃ 23 Lactate 0.7	
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Patient name: Miranda Ford **PROGRESS NOTES**

Date:
YESTERDAY

57yo
RC: Cough, SOB, fever
PMHx:
• Newly diagnosed myasthenia gravis 02/2019 that presented with fatigue, weakness, dysphagia, diplopia confirmed by EMG/+antibodies
• Found with anterior mediastinal mass (likely thymoma) to be resected in a few weeks
Rx: Mestinon 90mg QID
NKDA
Non-smoker/ ETOH socially
Lives with husband – one son who studies out of town

HPI:
Unwell x 1 week, with cough. Worse x 48 with SOB, fever. No infectious contact.
No other complaint.
Sat 88% RA – 93% on 4L/min - RR 20 – BP 95/60 – HR 110/min – T 38.5
Clinical examination unremarkable except for crackles right base.

Chest x-ray: RLL infiltrate

Bloodwork:
Hg 115 WBC 15.4 Plt 229
Creat 45 lytes normal
Lact normal
Liver enzymes normal

Admit to GIM
IV ceftriaxone/azythro – IV fluids
Continue Mestinon

Reviewed with sr. resident

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Patient name: MIRANDA FORD IMAGING REPORTS

CT SCAN
CHEST
2 MONTHS
AGO

There is a well-defined solid moderately enhancing lesion with a few non-enhancing necrotic areas within, is seen in the anterior mediastinum. No calcification cystic or fat component is seen within. The lesion is anteriorly abutting the right mid costal cartilages and the sternum. However no obvious bony erosion or chest wall involvement is noted. Postero-medially it is abutting the right atrium and the ascending aorta with maintained intervening fat planes, no infiltration to the adjacent structures is seen.

Chest x-ray
Yesterday

Right lower lobe infiltrate compatible with infection or atelectasis new compared to previous chest imaging.
Also visible is an enlarged mediastinum likely attributable to known anterior mediastinal mass on previous chest CT.

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Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Medical Management Questions (*see cheat sheet*):

- 1) How do we identify acute respiratory failure in neuromuscular weakness?
- 2) What is your approach to managing respiratory failure secondary to neuromuscular weakness?
- 3) What is your approach to airway management with an anterior mediastinal mass?
- 4) What do you think triggered the myasthenic crisis in this case? What are some other common triggers of myasthenic crisis for clinicians to be aware of?
- 5) Aside from supportive care, how do we treat myasthenic crisis?
- 6) What are the key points in medical error disclosure?

Crisis Resource Management Questions:

- 7) How did the team handle this complex case?
- 8) When managing a patient outside of a resuscitation setting, what are some risks and benefits of transporting the patient to another location vs managing immediately?

Cheat Sheet:

- 9) How do we identify acute respiratory failure in neuromuscular weakness?
 - a. Bulbar weakness is likely to be the first sign of impending respiratory failure, including jaw and tongue weakness, staccato speech, nasal voice, choking, and poor cough. Dyspnea when lying supine should also be a red flag. Subtle signs may include tachypnea, a quiet voice, or weak neck muscles (difficulty lifting head off the bed).
 - b. Bedside respiratory function tests have been found to be predictive of imminent respiratory failure in some neuromuscular conditions. Rather than waiting for significant respiratory distress, elective intubation is recommended when Vital Capacity < 15-20 mL/kg or Maximal Inspiratory Pressure < 30 cm H₂O.
 - c. Do not wait for changes in the ABG before initiating ventilatory support, as they are very late findings. These typically include hypoxemia followed by hypercapnia.
- 10) What is your approach to managing respiratory failure secondary to neuromuscular weakness?
 - a. NIPPV or intubation are both reasonable options.
 - b. NIPPV may avert intubation in some myasthenic patients and is reasonable to trial. However, pCO₂ > 45-50 is a predictor of NIPPV failure.
 - c. When intubating, do not use depolarizing agents. Avoid or use low doses of nondepolarizing agents, as they will cause prolonged paralysis. Induction agents may be used in typical doses.
- 11) What is your approach to airway management with an anterior mediastinal mass?
 - a. Anterior mediastinal masses can cause major airway and cardiovascular compromise, especially under general anesthesia. Airway compression may be exacerbated in the supine position. SVC compression may lead to cardiovascular collapse.
 - b. The safest option is likely to be awake fiberoptic intubation, as it preserves spontaneous ventilation and allows for assessment of the level of tracheobronchial obstruction.
 - c. Maintaining spontaneous ventilation and avoiding paralysis are recommended; light sedation with ketamine would be appropriate.

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- d. Patients may need to be induced in a semierect or full seated position; once intubated, proning may be required if there are significant compressive effects of the mass (i.e. hypoxia, persistent hypotension).
- e. If possible, you should aim to intubate distal to the level of tracheobronchial obstruction, which may necessitate mainstem intubation.

12) What do you think triggered the myasthenic crisis in this case? What are some other common triggers of myasthenic crisis for clinicians to be aware of?

- a. Any form of stress (emotional or physical) can trigger crisis. Infection triggers >30% of crises, most commonly respiratory infections. Aspiration pneumonitis is the second most common cause.
- b. Poor control of underlying disease, pregnancy, surgery, acute elevations in body temp, and emotional upset may all also be triggers.
- c. Many common medications may trigger myasthenic crisis; likely azithromycin (a macrolide antibiotic) in this case.

Table 4. Commonly Used Medications in the Emergency Department that May Worsen Myasthenia Gravis

Antimicrobials	Aminoglycosides Quinolones Macrolides Antimalarials
Anticonvulsants	Phenytoin Carbamazepine
Antipsychotics	Chlorpromazine Prochlorperazine
Cardiovascular agents	Beta-blockers (including topicals) Calcium channel blockers Class I antidysrhythmics (lidocaine and procainamide)
Neuromuscular-blocking agents	Succinylcholine
Muscle relaxants	Long-acting benzodiazepines Baclofen
Others	Iodinated radiocontrast agents Magnesium sulfate and citrate

Data from multiple sources (2,6,7,12,44,56,62).

13) Aside from supportive care, how do we treat myasthenic crisis?

- a. Plasmapheresis and IVIG are the mainstays of therapy for myasthenic crisis, although they both take days to weeks to reach peak effect, and therefore cannot be used to prevent the need for ventilatory support in a patient in respiratory failure.
- b. IgA deficiency screening is required prior to starting IVIG.
- c. Corticosteroids should be started after plasmapheresis or IVIG, as they may paradoxically worsen myasthenic crisis.
- d. Pyridostigmine is a maintenance therapy that should be stopped in crisis, as it can cause dysrhythmias and will worsen secretions.

14) What are the key points in medical error disclosure?

- a. The facts about what happened
- b. The steps that were and will be taken to minimize harm
- c. That you / the healthcare team are sorry for what happened
- d. What will be done to prevent similar events in future

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References

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